

Request for Information on the Development of an Artificial Intelligence (AI) Action Plan

Submitted by: José-Marie Griffiths, President, Dakota State University
and member of the National Security Commission on Artificial Intelligence

About Dakota State University

Dakota State University (DSU) is a STEM-focused institution emerging as a leader in cyber, with vast experience in AI education and workforce development. Our faculty, staff, and students are advancing the development, application and research boundaries of AI and other emerging technologies. DSU is pleased to contribute and participate in these efforts with our academic excellence, research work, and workforce development programs.

Dakota State Education Efforts with AI

At Dakota State, we work with AI on many fronts.

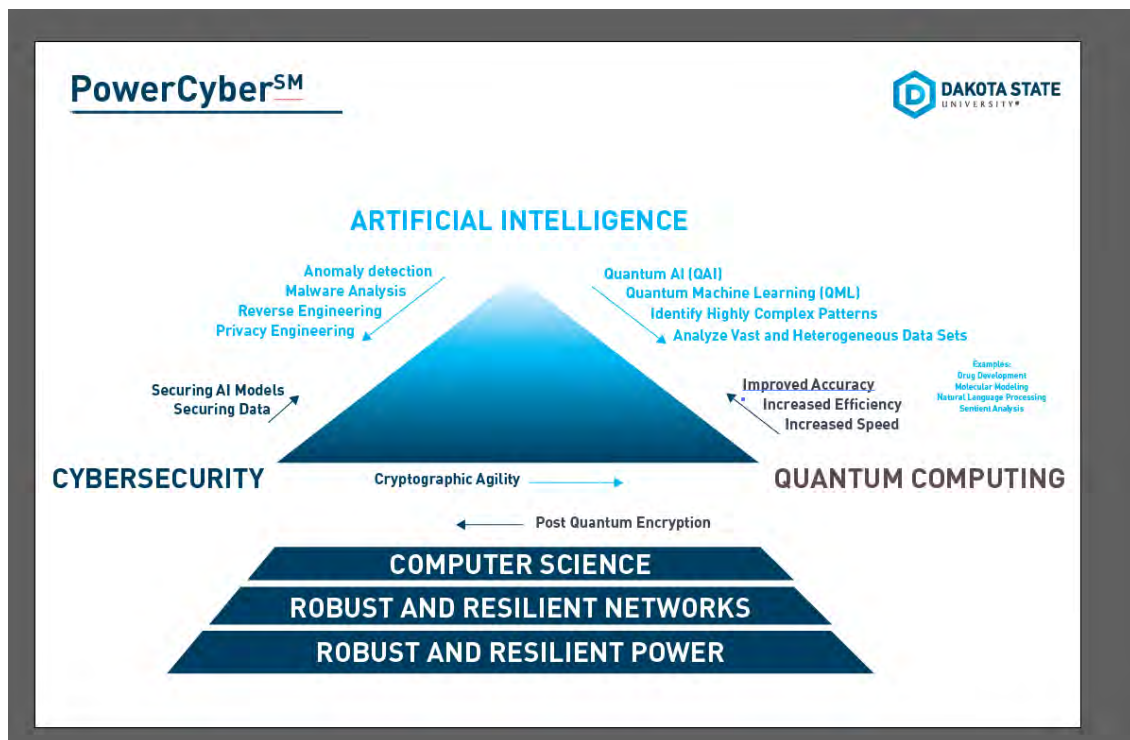
- Developing AI (The Beacom College of Computer and Cyber Sciences)
- Applying AI (The College of Business & Information Systems)
- Using AI (university-wide)

Recommendations for the AI Action Plan

- **AI Risk-Based Regulation:** The federal government should continue to legislate on AI related issues, in particular risks associated with AI. This approach would involve categorizing AI systems according to the potential risks or threats they pose, with higher levels of scrutiny for those systems that have the most significant societal impact, such as autonomous vehicles or AI in healthcare. A tiered regulation system would allow Congress to balance the need for oversight with the goal of not stifling technological progress. Therefore, encouraging responsible AI development and pleasing both sides of the AI regulation debate.
- **Public-Private Partnerships:** The federal government must encourage public-private partnerships to foster collaboration between government agencies, private companies, and research institutions such as higher education. These groups can share expertise, resources, and data to develop policies that promote ethical and safe AI innovation. Public-private partnerships would provide a framework for establishing industry standards and guidelines that promote transparency, fairness, and accountability, ensuring AI technologies benefit society as a whole.
- **Rural Networks:** Public-private partnerships should include rural networks, which could leverage collaborations for AI research and development as well as adoption. This would limit the potential for a digital divide in rural areas, such as the Northern Plains. An example is our partnership with a five-state consortium, to leverage generative AI to enhance research, education, and economic development in Idaho, Montana, North

Dakota, South Dakota, and Wyoming. This group will help with geographical and socio-economic disparities from the emergence of generative AI technologies, and ensure affordable, accessible, trustworthy AI at speed and scale in the rural states.

- **Developer's Transparency and Accountability:** To build public trust and ensure responsible AI development, the federal government should pass legislation that mandates transparency and accountability from companies, organizations, or institutions developing AI technologies. This would require companies to disclose key aspects of their AI models, certain data sources, decision-making processes, and potential biases. Accountability mechanisms must be put in place to hold companies responsible for the outcomes of their AI systems, ensuring that they are liable for harm caused by faulty or biased algorithms and are incentivized to prioritize ethical practices.
- **Data Privacy/Data Protection in AI:** Given the vast amounts of data relied on by AI systems, Congress must enact strong data privacy and protection laws that specifically address the use of data in AI. This includes ensuring that individual personal data is safeguarded against misuse and that AI developers comply with strict guidelines on data collection, consent, storage, and usage.
- **Higher Education Investments in AI Education and Training:** The federal government must prioritize investments in AI education and training programs to prepare the workforce of tomorrow for the future of AI. This includes funding for higher education institutions to develop specialized AI curriculum, as well as providing retraining opportunities for individuals whose jobs are at risk of automation. It could also include expanded funding for scholarship for service programs such as CyberCorps®. By investing in education, Congress will ensure the U.S. remains competitive in the global AI landscape and that workers are equipped with the skills needed to thrive in an AI driven economy.
- **K-12 Education Investments:** Educational investments should not only take place with higher education, but also at the K-12 level. Helping students explore technology at a young age means they can more fully access skills and opportunities needed for economic success in their adult years. An example of this is our dual-credit program, the Governors Cyber Academy, which provides high school juniors and seniors with a jumpstart in high-demand cyber career occupations.
- **Legal Immigration Efforts:** Because domestic talent will be insufficient on its own to fill the workforce gaps needed in technology fields, attention should be given to efforts such as the NSCAI's recommendation to remove caps on green cards for exceptional talent. Also, non-citizen graduates with cyber degrees from accredited universities should receive faster access to green cards.
- **The Power of AI Research Implementation:** Research and Development in Artificial Intelligence should be structured to ensure that innovation will not be isolated, but implemented in consideration with the concepts of cybersecurity and quantum computing. This concept is best visualized with Dakota State's Power CyberSM graphic.



DSU's Leading Degree Programs

The university offers 46 degrees in computer and cyber sciences, business and information systems, education and human performance and arts & sciences, including eight master's and four doctoral degrees. More degrees involving emerging technologies are in the planning stages. Current AI and quantum degrees include:

- Artificial Intelligence (MS)
- Analytics and Applied Artificial Intelligence (MS)
- Computer Science specialization in AI (MS)
- Artificial Intelligence (BS)
- AI in Organizations (BS)
- AI and Machine Learning minor
- Quantum computing for cybersecurity graduate certificate
- Quantum computing for cybersecurity minor

AI Research

Through Dakota State University's Madison Cyber Labs, the university is actively conducting applied research in numerous areas, from adaptive technology to AI and Quantum research.

We also have the Dakota State Applied Research Lab (DSU-ARL), which contracts with companies and agencies to work on highly technical projects including software development, malware analysis, digital forensics, cybersecurity, workforce development, and penetration

testing. Dakota State University's Applied Research Corporation (DARC) has been established as a non-profit corporation to develop, operate, and expand the DSU-ARL. One ARL building is currently operational in Madison, S.D.; a second building is under construction in Sioux Falls, S.D.

We have numerous established partnerships with businesses and organizations, from international collaborations with AI Sweden and the Edge Learning Consortium and Griffith University (Australia), international corporations such as Case New Holland and Google, along with many national defense contractors, as well as with state agencies and regional companies.

We are also on the cutting-edge of emerging technologies, having established the Center for Quantum Information Science and Technology (C-QIST), a partnership between Dakota State and the South Dakota School of Mines.

Research Projects:

- **AI in K-12 Education:**
 - **Deb Tech, Brenda Merkel, and David De Jong** led professional development for SD's K-12 business educators on **December 4th**, presenting on **disruptive technologies, business careers, and AI integration in classrooms**. Sponsored by **DSU and the South Dakota Department of Education**.
 - At the **2024 Rosebud Sioux Tribe Professionals Conference** (December 20-22), DSU faculty presented:
 - **"How AI Can Help You in Your Work"** – **Brenda Merkel and David Zeng** demonstrated how AI enhances productivity and efficiency.
 - **"How to Learn About AI"** – **Fenecia Homan, Austin O'Brien, and David De Jong** introduced attendees to **AI tools**, helping many **use AI for the first time**.
 - **Mark Spanier and Fenecia Homan** led **three professional development sessions** for **West Central School District educators** on January 20, covering:
 - **Governor's Cyber Academy opportunities**
 - **Foundational AI concepts**
 - **Classroom integration of AI tools**
- **AI Policy and Standards:**
 - **Omar El-Gayar** serves on the **ABET-CAC Criteria Committee**, co-chairing subcommittees focused on **developing AI/ML program criteria** and integrating AI into accreditation standards.
- **AI in Cybersecurity and Industry Collaboration:**
 - **Shawn Zwach, Kyle Cronin, and Fenecia Homan** presented **"Spurring Awareness and Access Through Cyber Enrichment Programming"** at the **2024 NICE K12 Cybersecurity Education Conference** in San Antonio, Texas.

AI Sweden Partnership & Paulson Center Expansion

- **AI Sweden Industrial Immersion & Exchange Program**

- **Now a key initiative under the Paulson Center for Cyber and Economic Development**, the **AI Sweden partnership** has expanded into a **year-round program** for AI and cybersecurity students.
- **In 2024, 23 students (12 from DSU, 11 from Swedish universities)** worked on **real-world AI & cybersecurity challenges** from **Volvo, Hewlett Packard Enterprise, Swedish Defence University, Case New Holland, and others**.
- The **10-week program** is split between **AI Sweden's Edge Learning Lab in Gothenburg, Sweden, and DSU's MadLabs in Madison, SD**.
- **Example projects include:**
 - **Poisoning Attacks on Federated Learning for Autonomous Driving** (Volvo, Zenseact)
 - **Intrusion Detection System for CAN Bus Networks** (Case New Holland)
 - **Development of AI-powered Honeypots for Cybersecurity** (Volvo, Västra Götalandsregionen)
 - **Transformer-Based Anomaly Detection for Maritime Security** (Swedish Defence University, Swedish Navy)
- **More details on AI Sweden & DSU's research collaboration:**
 - [2024 Projects](#)
 - [2023 Projects](#)
- **Dr. Mark Spanier and Jared Soundy** were identified as DSU's **AI integration leaders**, working on further expanding AI initiatives at DSU.

Enhancing Student Success

- **AI for Teacher Apprentices:**
 - **David De Jong** trained **Little Wound teacher apprentices and school leaders** on using **ChatGPT, MagicSchoolAI, and NotebookLM** in the classroom.
- **AI Research and Publications:**
 - DSU faculty and students published several **AI-related research papers**, including:
 - **"The Influence of Organizational Culture on AI Adoption and Organizational Performance"** – Mutale W. & Omar El-Gayar (Decision Sciences Institute Annual Conference)
 - **"A Review of Reasoning in Artificial Agents Using Large Language Models"** – Naidu, N. & El-Gayar, O. (HICSS)
 - **"Human Cognitive Bias Mitigation Approaches to Fairness in the Machine Learning Value Chain"** – Surles, S., Noteboom, C., & El-Gayar, O. (HICSS)

Advancing AI Research and Thought Leadership

- **NSF & AI Research Contributions:**
 - **Omar El-Gayar** was invited as a **reviewer and panelist for two NSF programs**, evaluating AI research proposals.

- o **DSU's first NSF I-Corps cohort** launched, focusing on **AI-driven entrepreneurship training** in collaboration with SD Mines. **Mike Roach and Andy Behrens** are leading the program.
 - o **Andy Behrens** serves as PI on a funded project with **Hetherington Group (HG)**, developing an **eight-week beginner-level data analytics course** using AI tools.
- **Cybersecurity & AI:**
 - o At the **IEEE Annual Computing and Communication Workshop and Conference**, DSU faculty and students won **Best Paper in Security, Trust, and Privacy** for work on **Quantum Security**, presented by **Ashay Kappala**.
- **Artificial Intelligence and Medical Ethics**
 - o **Pete Hoelsing** is contributing to an AI Ethics working group among the USD Sanford School of Medicine's Ethics and Humanities Faculty

AI Workforce Development

Dakota State was a pilot site for the initial GenCyber Camps in 2013. We currently host camps for teachers, middle school girls, high school co-ed, and are starting a camp in 2025 for middle school boys. The high school and girls camps are the largest residential sites in the nation for this program. This program is designed to inspire the next generation of the cyber workforce through hands-on exploration of technology concepts including digital forensics, cryptography, privacy, cyber awareness, and coding.

With support from the State of South Dakota, we have created the Governors Cyber Academy. This dual-credit program is designed to expand workforce development awareness by providing access to cyber education and career pathways for high school students throughout the state of South Dakota and beyond.

DSU-SOAR is a non-profit 501(c)(3) organization affiliated with Dakota State University, created to provide relevant workforce training needs for individuals, corporations, and government agencies.

we DSU works with AI, Cybersecurity, data privacy in:

- 1) National security, Intelligence, and defense
- 2) Critical infrastructure, such as healthcare and agriculture.

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Submitted by:

José-Marie Griffiths, President
 Dakota State University
 820 N Washington Ave
 Madison, SD 57042